

PCA on TONA Swaps

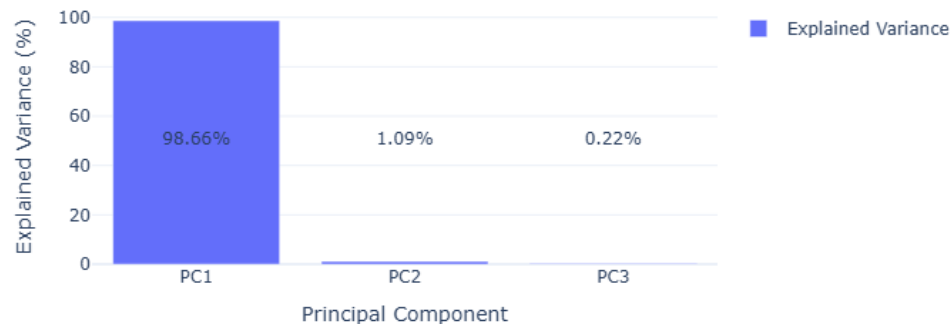
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Maths and Methodology

Explained variance by the first 3 principal components



Principal components against tenors



What is PCA?

PCA is a method that allows for the reduction of dimensionality in data

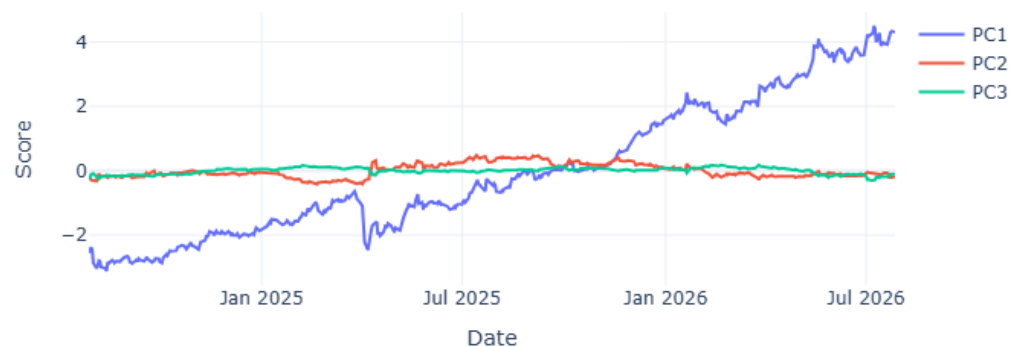
Method:

1. Centralise our data
2. Calculate the covariance matrix of the dataset and its eigenvectors (the PCs) that explain the majority of variance
3. Reconstruct our dataset

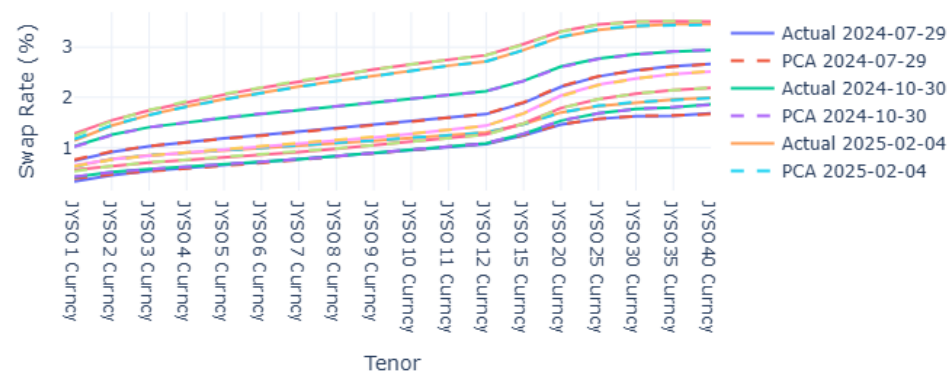
How does it apply to swaps?

- The principal components capture the main ways the swap curve moves: level, slope and curvature
- Using these component, we reconstruct a PCA-implied 'fair value' swap curve that reflects the curve's underlying structure

PCA factor scores over time



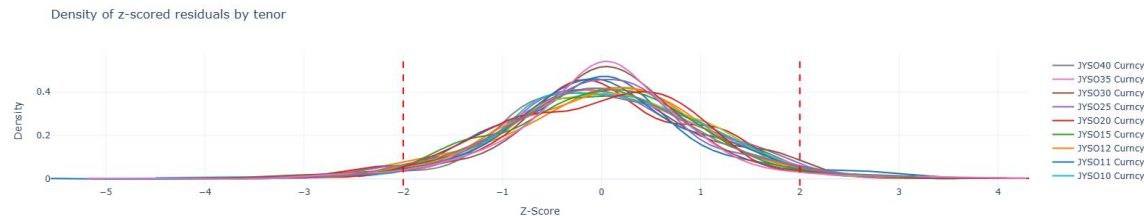
Actual swap curve vs PCA swap curve



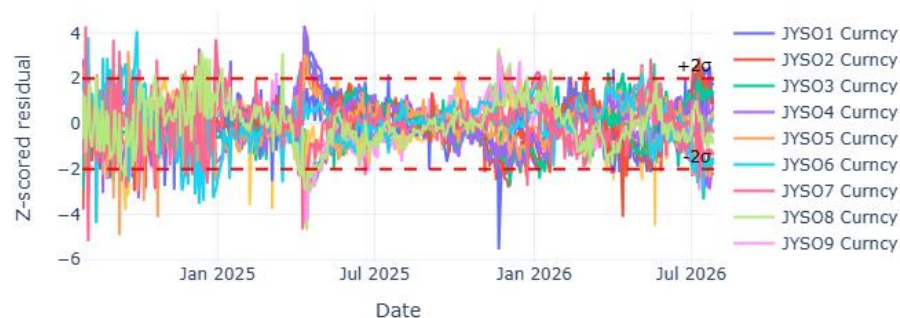
Residuals as a Trading Signal

Residuals as a signal:

- Residuals between the actual swap curve and PCA-reconstructed swap curve are seen as random and mean reverting
- A positive residual (market rate > model rate) suggests a swap is trading cheap
- A negative residual (market rate < model rate) suggests a swap is trading rich



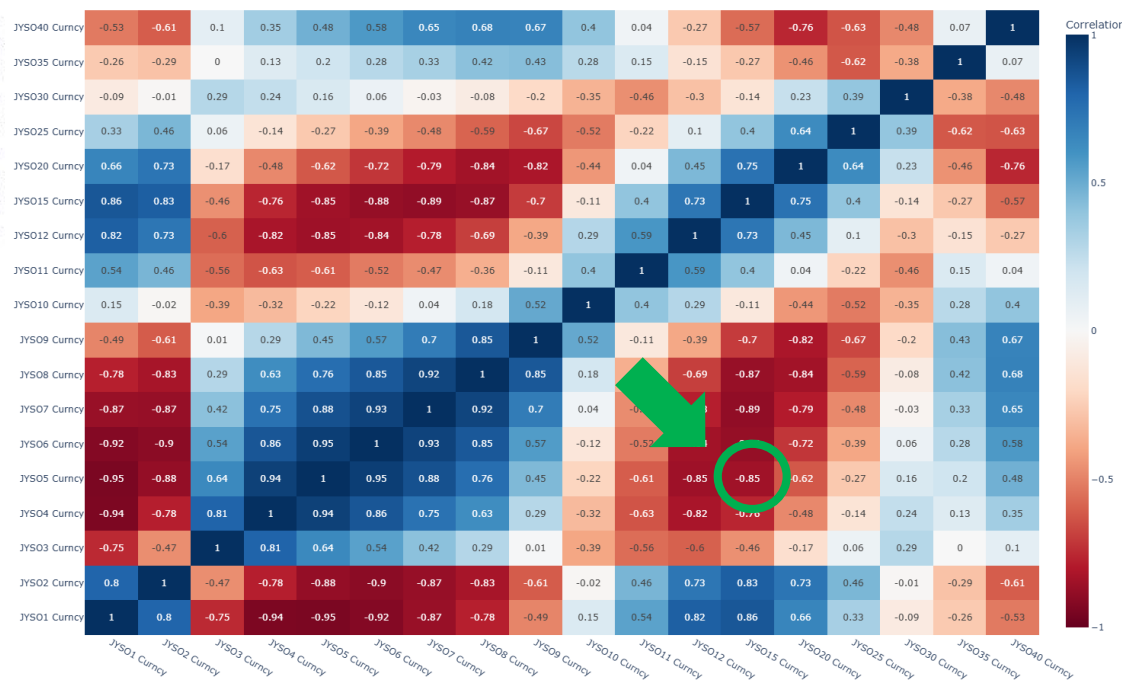
Z-scored residuals over time



Z-scored residuals over time



Correlations between the residuals of the swap rates

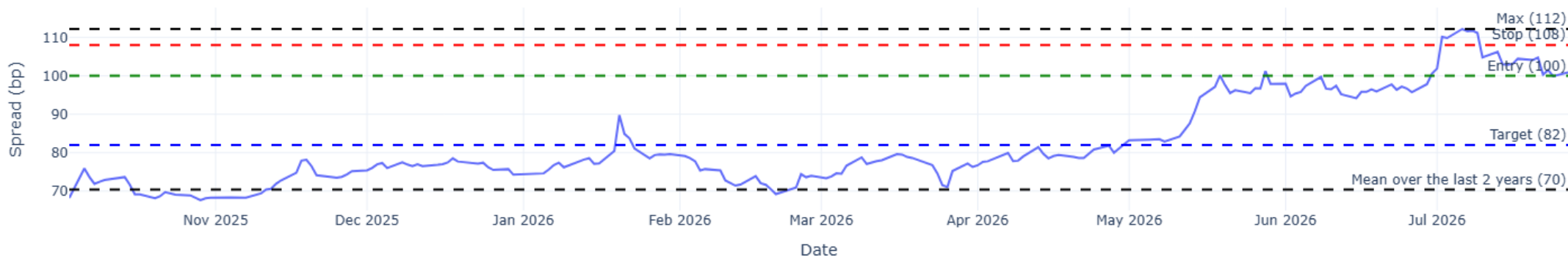


Trade Idea and Risks

5s15s Flattener:

Entry – 100bp, Target – 82bp, Stop loss – 108bp, Timeframe – 3-5 Months

Short JPY 15s5s Spread Trade



Rationale:

1. The 15yr is cheap with respect to the model and the 5 year is expensive with respect to the model, we aim to exploit a statistical inefficiency
2. Our target is about one standard deviation above the mean and below the entry
3. Timeframe of 3-5 months to see reversion

Risks:

1. Current fears of inflation, due to weak Yen (even though CPI came in at 1.7% below target)
2. Our signal may not persist if inputs change (period of data, number of swaps etc) as PC's are dependent on data used

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